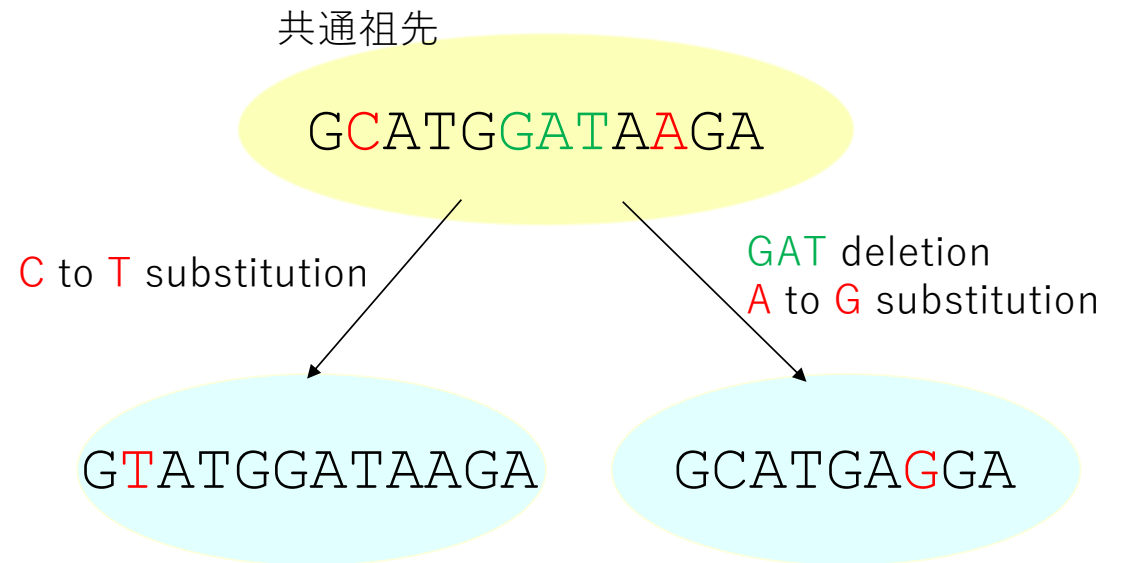
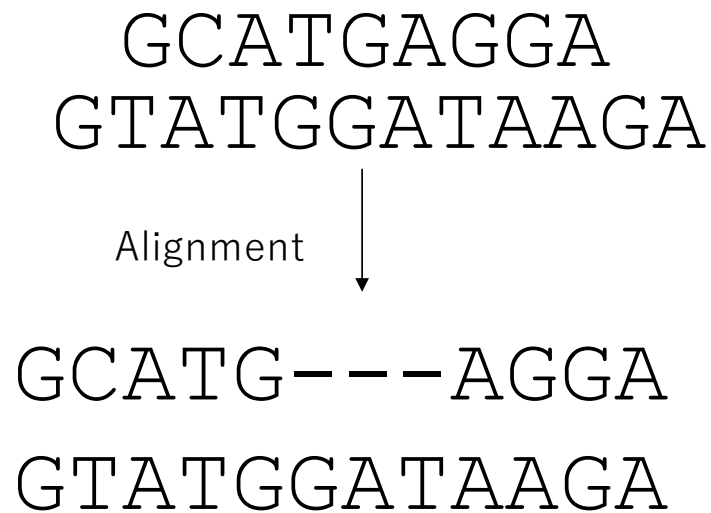


バイオインフォマティクス
基本ツール
(ホモロジー検索)

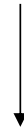
配列比較とアライメント



相同性 (homology) : 共通祖先の同一の構造から派生した構造であること
相同性は類似性 (similarity) に基づいて推定する

類似性スコアと配列アライメント

GCATGAGGA
GTATGGATAAGA



Score definition: match **+2**、mismatch **-1**、gap **-2**

GCATG---AGGA

GTATGGATAAGA

+2**-1****+2****+2****+2****-2****-2****-2****+2****-1****+2****+2** = **+6**

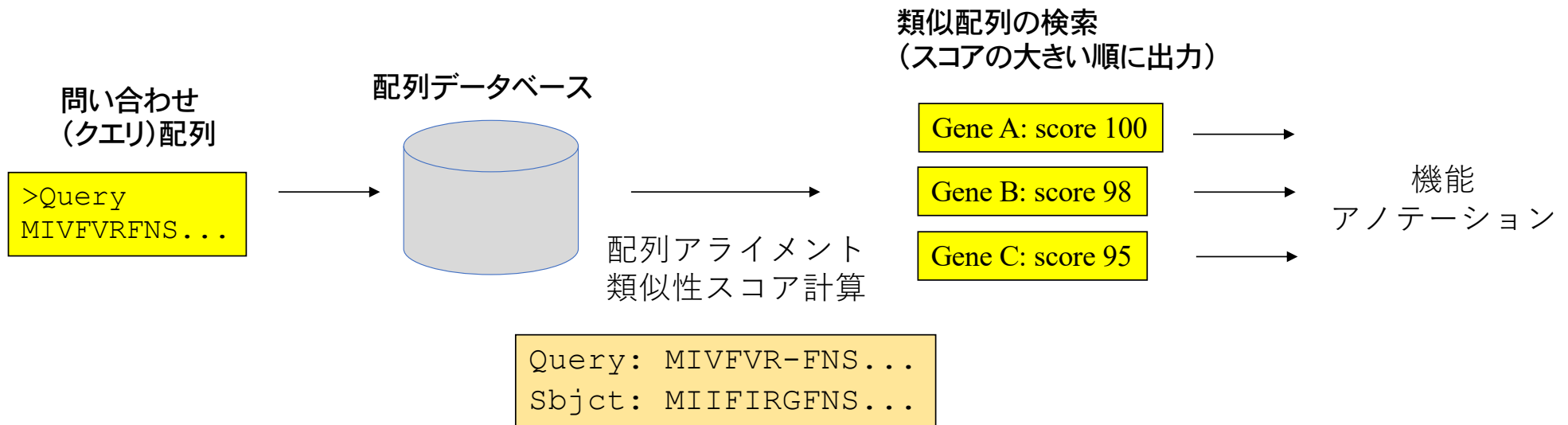
配列アライメント：類似性スコアが最大になる
ように各配列にギャップを挿入して並べる

アミノ酸の類似性行列：BLOSUM62

P	7																			
A	-1	4	small																	
G	-2	0	6																	
N	-2	-2	0	6	amide															
Q	-1	-1	-2	0	5															
D	-1	-2	-1	1	0	6	acidic													
E	-1	-1	-2	0	2	2	5													
T	-1	0	-2	0	-1	-1	-1	5	hydroxyl											
S	-1	1	0	1	0	0	0	1	4											
C	-3	0	-3	-3	-3	-3	-4	-1	-1	9										
V	-2	0	-3	-3	-2	-3	-2	0	-2	-1	4									
I	-3	-1	-4	-3	-3	-3	-3	-1	-2	-1	3	4	aliphatic (V,I,L)							
M	-2	-1	-3	-2	0	-3	-2	-1	-1	-1	1	1	5							
L	-3	-1	-4	-3	-2	-4	-3	-1	-2	-1	1	2	2	4						
K	-1	-1	-2	0	1	-1	1	-1	0	-3	-2	-3	-1	-2	5	basic				
R	-2	-1	-2	0	1	-2	0	-1	-1	-3	-3	-3	-1	-2	2	5				
H	-2	-2	-2	1	0	-1	0	-2	-1	-3	-3	-3	-2	-3	-1	0	8			
F	-4	-2	-3	-3	-3	-3	-3	-2	-2	-2	-1	0	0	0	-3	-3	-1	6	aromatic	
Y	-3	-2	-3	-2	-1	-3	-2	-2	-2	-2	-1	-1	-1	-1	-2	-2	2	3	7	
W	-4	-3	-2	-4	-2	-4	-3	-2	-3	-2	-3	-3	-1	-2	-3	-3	-2	1	2	11
	P	A	G	N	Q	D	E	T	S	C	V	I	M	L	K	R	H	F	Y	W

Higher score is assigned between similar amino acids

類似性（相同性）検索



問い合わせ配列の準備

FASTA形式

```
>エントリー名 説明  
配列.....  
>エントリー名 説明  
配列.....  
.....
```

```
>gi|4758884|ref|NP_004553.1| parkin isoform 1 [Homo sapiens]  
MIVFVRFNSSHGFPVEVDSITSIFQLKEVVAKRQGV PADQLRVIFAGKELRNDWTVQNC DLDQQSIVHIV  
QRPWRKGQEMNATGGDDPRNAAGGCERE PQSLTRVDLSSSVLP GDSVGLAVILHTDSRKDSPPAGSPAGR  
SIYNSFYVYCKGPCQRVQPGKLRVQCSTCRQATLT LTQG PSCWDDVLIPNRMSGECQSPHCPGTSAEFFF  
KCGAHPTSDKETPVALH LIATNSRNITCITCTDVRSPVLV FQCNSRHVICLD CFHLYCVTRLNDRQFVHD  
PQLGYSLPCVAGCPNSLIKELHHFRILGEEQYNRYQQYGAEECVLQMGGVLCPRPGCGAGLLPEPDQRKV  
TCEGGNGLGCGFAFCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTTKPCPR  
CHVPVEKNGGCMHMKCPQPQCRLEWCWNCGEWNRVCMGDHWF DV
```

BLAST (Basic Local Alignment Search Tool)

The screenshot displays the BLAST web interface. At the top, there's a navigation bar with links: Home, Recent Results, Saved Strategies, and Help. Below this, the main heading is "Basic Local Alignment Search Tool". A brief description states: "BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance." A "Learn more" link is provided. To the right, a "NEWS" section dated "Mon, 21 Jul 2025" highlights a new release: "Download BLAST+ 2.17.0 now!" with a "More BLAST news..." link. The "Web BLAST" section features three main options: "Nucleotide BLAST" (nucleotide > nucleotide), "blastx" (translated nucleotide > protein), and "tblastn" (protein > translated nucleotide). Below these, there's a "Protein BLAST" option (protein > protein). A "BLAST Genomes" section includes a search bar for "Enter organism common name, scientific name, or tax id" with a "Search" button and links for Human, Mouse, Rat, and Microbes. The "Standalone and API BLAST" section offers three ways to use BLAST: "Download BLAST" (Get BLAST databases and executables), "Use BLAST API" (Call BLAST from your application), and "Use BLAST in the cloud" (Start an instance at a cloud provider). Finally, the "Specialized searches" section lists eight tools: SmartBLAST (Find proteins highly similar to your query), Primer-BLAST (Design primers specific to your PCR template), Global Align (Compare two sequences across their entire span (Needleman-Wunsch)), CD-search (Find conserved domains in your sequence), IgBLAST (Search immunoglobulins and T cell receptor sequences), VecScreen (Search sequences for vector contamination), CDART (Find sequences with similar conserved domain architecture), and Multiple Alignment (Align sequences using domain and protein constraints).

program	query	database	translation
BLASTP	protein	protein	
BLASTN	nucleotide	nucleotide	
BLASTX	nucleotide	protein	query
TBLASTN	protein	nucleotide	database
TBLASTX	nucleotide	nucleotide	both

- 核酸配列は、相補鎖も同時に検索する
- 翻訳は、各鎖の3つの読み枠、計6通りの読み枠で比較する

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>

BLAST 検索インターフェイス

クエリ配列

BLAST[®] » blastp suite

blastn **blastp** blastx tblastn tblastx

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

>gi|4758884|ref|NP_004553.1| parkin isoform 1 [Homo sapiens]
MIVFVRFNSSHGFPVEVSDTSIFQLKEVAKRQGVADQLRVIFAGKELRNDW
TVQNCDLQQSIVHIV
QRPWRKGQEMNATGGDDPRNAAGGCEREPSLTRVDLSSSVLPGDSVGLAVI
LHTDSRKDSPSPAGSPAGR
STTNSP...VYCKGPCQQRVQPGKLRVQCSTCRQATLTLTQGPSCWDDVLIPNRMS
GECQSPHCPGTSAEFF
KCGAHTSDKETPVALHLIATNSRNITCITCTDVRSPVLVFCNSRHVICLDCFHL
YCVTRLNDRQFVHD
PQLGYSLPCVAGCPNSLIKELHHFRILGEEQYNRYQQYGAECEVLQMGGVLC
RPGCGAGLLPEPDQRKV
TCEGGNGLGCGFAFCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQAR
WEAASKETIKKTKPCPR
CHVPVEKNGGCMHMKCPQPCRLWCWNCGEWNRVCMGDHWFVDV

Query subrange ?
From
To

Or, upload file ?

Job Title
Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database ?

Organism Optional
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. ?

Program Selection

Algorithm ☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
Choose a BLAST algorithm ?

BLAST Search database ClusteredNR using Blastp (protein-protein BLAST)
☐ Show results in a new window

データベース

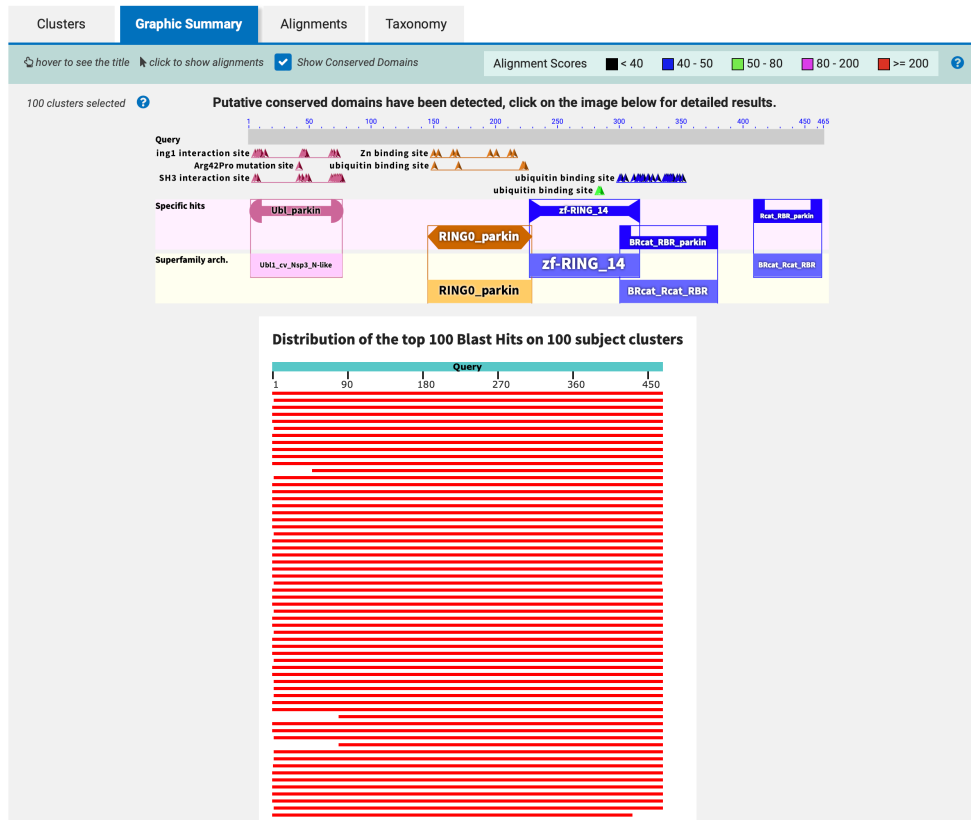
BLAST 検索結果

Clusters		Graphic Summary		Alignments		Taxonomy													
Clusters producing significant alignments						Download		Select columns		Show		100							
☒ select all 100 clusters selected						GenPept		Graphics		Distance tree of results		Multiple alignment		MSA Viewer					
Cluster Composition		Cluster Ancestor		Cluster Representative Sequence		Max Score		Total Score		Query Cover		E value		Per. Ident		Acc. Len		Accession	
Click the  to see the cluster contents																			
☒	1 member(s), 1 organism(s)	other sequences		fusion protein HA-miRFP670nano-Parkin [synthetic construct]		975		975		100%		0.0		100.00%		627		BET92570.1	
☒	4 member(s), 4 organism(s)	primates		E3 ubiquitin-protein ligase parkin isoform X10 [Macaca nemestr...		940		940		100%		0.0		97.19%		530		XP_070952036.1	
☒	1 member(s), 1 organism(s)	white-tufted-ear marmoset		E3 ubiquitin-protein ligase parkin isoform X2 [Callithrix jacchus]		929		929		100%		0.0		95.70%		489		XP_078226555.1	
☒	82 member(s), 26 organism(s)	primates		E3 ubiquitin-protein ligase parkin isoform 2 [Homo sapiens]		894		894		100%		0.0		93.98%		437		NP_054642.2	
☒	8 member(s), 7 organism(s)	prosimians		E3 ubiquitin-protein ligase parkin isoform 1 [Daubentonia mada...		870		870		100%		0.0		89.48%		465		KAL2792878.1	
☒	6 member(s), 1 organism(s)	Philippine flying lemur		E3 ubiquitin-protein ligase parkin isoform X4 [Cynocephalus vol...		858		858		100%		0.0		88.55%		480		XP_062952603.1	
☒	30 member(s), 18 organism(s)	even-toed ungulates & wh...		E3 ubiquitin-protein ligase parkin isoform X2 [Hippopotamus am...		854		854		100%		0.0		87.31%		495		XP_057594137.1	
☒	12 member(s), 6 organism(s)	odd-toed ungulates		E3 ubiquitin-protein ligase parkin isoform X1 [Equus quagga]		853		853		100%		0.0		87.34%		466		XP_046526636.1	
☒	2 member(s), 1 organism(s)	Chinese tree shrew		E3 ubiquitin-protein ligase parkin isoform X2 [Tupaia chinensis]		847		847		100%		0.0		87.74%		464		XP_006152316.1	
☒	1 member(s), 1 organism(s)	puma		E3 ubiquitin-protein ligase parkin [Puma concolor]		843		843		100%		0.0		86.24%		691		XP_025781231.1	
☒	1 member(s), 1 organism(s)	gray mouse lemur		E3 ubiquitin-protein ligase parkin isoform X2 [Microcebus murin...		842		842		100%		0.0		88.60%		464		XP_075858924.1	
☒	9 member(s), 4 organism(s)	primates		E3 ubiquitin-protein ligase parkin isoform X2 [Symphalangus sy...		839		839		90%		0.0		95.93%		442		XP_055125023.2	
☒	48 member(s), 23 organism(s)	carnivores		E3 ubiquitin-protein ligase parkin isoform X1 [Phoca vitulina]		838		838		100%		0.0		86.39%		469		XP_032263409.1	
☒	2 member(s), 1 organism(s)	gray squirrel		E3 ubiquitin-protein ligase parkin isoform X2 [Sciurus carolinen...		838		838		100%		0.0		86.67%		462		XP_047413015.1	
☒	1 member(s), 1 organism(s)	American beaver		E3 ubiquitin-protein ligase parkin isoform X1 [Castor canadensis]		836		836		100%		0.0		80.39%		509		XP_073926901.1	
☒	5 member(s), 2 organism(s)	elephants		E3 ubiquitin-protein ligase parkin isoform X1 [Elephas maximus...		833		833		100%		0.0		85.16%		465		XP_049762059.1	
☒	26 member(s), 14 organism(s)	cat family		E3 ubiquitin-protein ligase parkin isoform X1 [Felis catus]		830		830		100%		0.0		83.26%		481		XP_044914560.1	
☒	5 member(s), 2 organism(s)	pigs		E3 ubiquitin-protein ligase parkin [Sus scrofa]		830		830		100%		0.0		86.02%		461		NP_001038068.2	
☒	11 member(s), 4 organism(s)	even-toed ungulates & wh...		E3 ubiquitin-protein ligase parkin isoform X2 [Camelus ferus]		828		828		100%		0.0		86.02%		460		XP_032341198.1	
☒	2 member(s), 1 organism(s)	prairie deer mouse		E3 ubiquitin-protein ligase parkin isoform X2 [Peromyscus mani...		825		825		100%		0.0		82.24%		489		XP_076408996.1	
☒	1 member(s), 1 organism(s)	aardvark		E3 ubiquitin-protein ligase parkin [Orycteropus afer afer]		825		825		100%		0.0		84.23%		467		XP_042637507.1	
☒	5 member(s), 1 organism(s)	nine-banded armadillo		E3 ubiquitin-protein ligase parkin isoform X5 [Dasypus novemci...		823		823		100%		0.0		84.30%		464		XP_058145644.1	
☒	2 member(s), 2 organism(s)	rabbits & hares		E3 ubiquitin-protein ligase parkin isoform X1 [Ochotona princeps]		823		823		100%		0.0		84.95%		464		XP_058512164.1	
☒	58 member(s), 22 organism(s)	rodents		E3 ubiquitin-protein ligase parkin isoform 3 [Mus musculus]		820		820		100%		0.0		83.87%		465		NP_001390399.1	
☒	4 member(s), 3 organism(s)	dog, coyote, wolf, fox		E3 ubiquitin-protein ligase parkin isoform X1 [Nyctereutes procy...		818		818		100%		0.0		81.95%		481		XP_055168029.1	
☒	1 member(s), 1 organism(s)	Arctic ground squirrel		E3 ubiquitin-protein ligase parkin isoform 1-T1 [Urocitellus parryi]		818		818		100%		0.0		82.85%		482		KAM4816522.1	
☒	1 member(s), 1 organism(s)	golden spiny mouse		E3 ubiquitin-protein ligase parkin isoform X3 [Acomys russatus]		817		817		100%		0.0		83.44%		465		XP_051020298.1	
☒	3 member(s), 1 organism(s)	Nile rat		E3 ubiquitin-protein ligase parkin isoform X3 [Arvicanthis niloticus]		815		815		99%		0.0		84.63%		461		XP_076782759.1	
☒	9 member(s), 3 organism(s)	bats		E3 ubiquitin-protein ligase parkin isoform X1 [Rhinolophus ferru...		814		814		100%		0.0		83.44%		463		XP_032959339.1	

E-value
ランダムな配列で、そのスコア以上の類似性を持つ個数の期待値

BLAST 検索結果

グラフィカル表示



アライメント

Clusters Graphic Summary Alignments Taxonomy

Alignment view Pairwise Restore defaults Download

100 clusters selected

Download GenPept Graphics Next Previous Descriptions

fusion protein HA-miRFP670nano-Parkin [synthetic construct]

Sequence ID: [BET92570.1](#) Length: 627 Number of Matches: 1

Range 1: 163 to 627 GenPept Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
975 bits(2520)	0.0	Compositional matrix adjust.	465/465(100%)	465/465(100%)	0/465(0%)

Query	1	MIVFVRFNSSHGFPVEVSDSTIFQLKEVAKRQVADQLRVIFAGKELRNDWTVQNC	60
Sbjct	163	MIVFVRFNSSHGFPVEVSDSTIFQLKEVAKRQVADQLRVIFAGKELRNDWTVQNC	222
Query	61	LDQOSIVHIVQRPWRKGOENMATGGDDPRNAAGGCEREPOSLTRVDLSSSVLPDGSVGLA	120
Sbjct	223	LDQOSIVHIVQRPWRKGOENMATGGDDPRNAAGGCEREPOSLTRVDLSSSVLPDGSVGLA	282
Query	121	VILHTDSRKDSPAGSPAGRSIYNSFYVYCKGPCORVOPGKLRVQCSTCRQATLTLTQGP	180
Sbjct	283	VILHTDSRKDSPAGSPAGRSIYNSFYVYCKGPCORVOPGKLRVQCSTCRQATLTLTQGP	342
Query	181	SCWDDVLIPINRMSGECOSPHCPGTSAEFFFKCGAHTSDKETSVALHLIATNSRNITCIT	240
Sbjct	343	SCWDDVLIPINRMSGECOSPHCPGTSAEFFFKCGAHTSDKETSVALHLIATNSRNITCIT	402
Query	241	CTDVRSPVLVFQNSRHVICLDCFHYCVTRLNDRQFVHDPQLGYSLPCVAGCPNSLIKE	300
Sbjct	403	CTDVRSPVLVFQNSRHVICLDCFHYCVTRLNDRQFVHDPQLGYSLPCVAGCPNSLIKE	462
Query	301	LHFIRILGEEQYNYQYGAEECVLMGGVLCPRPGCGAGLLPEPDQRKVTCEGGNGLGC	360
Sbjct	463	LHFIRILGEEQYNYQYGAEECVLMGGVLCPRPGCGAGLLPEPDQRKVTCEGGNGLGC	522
Query	361	GFAPCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTKPCPR	420
Sbjct	523	GFAPCRECKEAYHEGECSAVFEASGTTTQAYRVDERAAEQARWEAASKETIKKTKPCPR	582
Query	421	CHVPVEKNGGCHMKCPQPCRLWCNCGCEWNRVCMGDHWFVDV	465
Sbjct	583	CHVPVEKNGGCHMKCPQPCRLWCNCGCEWNRVCMGDHWFVDV	627

Download GenPept Graphics Next Previous Descriptions

E3 ubiquitin-protein ligase parkin isoform X10 [Macaca nemestrina]

Sequence ID: [XP_070952036.1](#) Length: 530 Number of Matches: 1

Range 1: 68 to 530 GenPept Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
940 bits(2430)	0.0	Compositional matrix adjust.	450/463(97%)	453/463(97%)	0/463(0%)

コマンドライン BLAST

プログラム名 -query クエリ配列ファイル -db データベース名

プログラム名： blastp, blastn, tblastn, blastx, tblastx

- SwissProt データベースに対して検索を実行する。
CPUコア 4つを使って実行する (-num_threads)。
% blastp -query query.fa -db swissprot -num_threads 4 > blastout
- 同じ検索を行い、テーブル形式で出力する (-outfmt 6)。
% blastp -query query.fa -db swissprot -num_threads 4 -outfmt 6 > blasttab

BLASTの出力

BLASTP 2.16.0+

Database: SWISS-PROT: SWISS-PROT protein sequence database Release 2025_04
573,661 sequences; 207,922,125 total letters

Query= gi|4758884|ref|NP_004553.1| parkin isoform 1 [Homo sapiens]

Length=465

Sequences producing significant alignments:

		Score	E
		(Bits)	Value
O60260	[PRKN_HUMAN] RecName: Full=E3 ubiquitin-protein ligase par...	964	0.0
Q9JK66	[PRKN_RAT] RecName: Full=E3 ubiquitin-protein ligase parki...	830	0.0
Q9WVS6	[PRKN_MOUSE] RecName: Full=E3 ubiquitin-protein ligase par...	812	0.0
Q7KTX7	[PRKN_DROME] RecName: Full=E3 ubiquitin-protein ligase par...	382	7e-128
E0VIU9	[PRKN_PEDHC] RecName: Full=E3 ubiquitin-protein ligase par...	344	3e-113
Q6T486	[RBRA_DICDI] RecName: Full=Probable E3 ubiquitin-protein l...	77.8	5e-14
Q84RR0	[ARI7_ARATH] RecName: Full=Probable E3 ubiquitin-protein l...	77.0	9e-14
Q6PFJ9	[ARI1_DANRE] RecName: Full=E3 ubiquitin-protein ligase ari...	70.5	1e-11
Q9Z1K5	[ARI1_MOUSE] RecName: Full=E3 ubiquitin-protein ligase ARI...	70.5	1e-11
A2VEA3	[ARI1_BOVIN] RecName: Full=E3 ubiquitin-protein ligase ARI...	70.5	1e-11
Q9Y4X5	[ARI1_HUMAN] RecName: Full=E3 ubiquitin-protein ligase ARI...	70.5	1e-11
Q94981	[ARI1_DROME] RecName: Full=E3 ubiquitin-protein ligase ari...	70.1	1e-11
Q6NW85	[ARI1L_DANRE] RecName: Full=E3 ubiquitin-protein ligase ar...	70.1	1e-11
B1H1E4	[ARI1_XENTR] RecName: Full=E3 ubiquitin-protein ligase ari...	69.7	2e-11
Q32NS4	[ARI1_XENLA] RecName: Full=E3 ubiquitin-protein ligase ari...	68.6	4e-11
Q8L829	[ARI5_ARATH] RecName: Full=Probable E3 ubiquitin-protein l...	67.0	1e-10
Q9UBS8	[RNF14_HUMAN] RecName: Full=E3 ubiquitin-protein ligase RN...	65.5	3e-10
Q9Z1K6	[ARI2_MOUSE] RecName: Full=E3 ubiquitin-protein ligase ARI...	65.1	5e-10
P0C8K8	[ARI6_ARATH] RecName: Full=Putative E3 ubiquitin-protein l...	64.7	8e-10
O95376	[ARI2_HUMAN] RecName: Full=E3 ubiquitin-protein ligase ARI...	64.7	8e-10
Q9SKC3	[ARI9_ARATH] RecName: Full=Probable E3 ubiquitin-protein l...	63.5	2e-09
Q9JI90	[RNF14_MOUSE] RecName: Full=E3 ubiquitin-protein ligase RN...	61.6	6e-09
F1RB95	[RNF14_DANRE] RecName: Full=E3 ubiquitin-protein ligase RN...	61.2	9e-09
Q05120	[UBIL_NPVOP] RecName: Full=Ubiquitin-like protein; Flags: ...	55.5	1e-08

>Q7KTX7 [PRKN_DROME] RecName: Full=E3 ubiquitin-protein ligase
parkin {ECO:0000303|PubMed:12642658}; EC=2.3.2.31 {ECO:0000269|PubMed:17456438,
ECO:0000269|PubMed:20194754, ECO:0000269|PubMed:23770917,
ECO:0000269|PubMed:24901221};
Length=482

Score = 382 bits (981), Expect = 7e-128, Method: Compositional matrix adjust.
Identities = 200/474 (42%), Positives = 280/474 (59%), Gaps = 33/474 (7%)

Query	1	MIVFVRFNSSHGFPVEVSDTSIFQLKEVVAKRQGVFADQLRVIFAGKELRNDWTVQNCD	60
		+ ++V+ N+ V ++ I +KE+VA + G+ D L++IFAGKEL + T++ CD	
Sbjct	30	LSIYVKNTNTGKTLTVNLEFPQWDIKNVKELVAPQLGLQPDDLKIIIFAGKELSDATTIEQCD	89

Query	61	LDQQSIVHIVQ-RPWRKGQEMNATGGDDPRNAAGGCEREPPQSLTRVDLSSSVLPGDSVGL	119
		L QQS++H ++ RP + Q++ + ++ + +P + T +DL	
Sbjct	90	LGQQSVLHAIRLRPPVQRQKIQSATLEEEEPSSLDEASKPLNETLLDLQ-----	138

Query	120	AVILHTDSRKDSPPAGSPAGRSIYNSFYVYCKGFCQVRVQPGKLRVQCSTCRQATLTLTQG	179
		L ++ R + ++ F+V+C C ++ GKLRV+C+ C+ T+ +	
Sbjct	139	---LESEERLNITDEERVRAKA---HFFVHC-SQCDKLCNGKLRVRCALCKGGAFTVHRD	191

Query	180	PSCWDDVLIPNRMSGECQSPHCPGTS-----AEFFFKCGAHPTS-DKETPVALHLIA	230
		P CWDDVL R+ G C+S AEFFFKC H + +K+ L+LI	
Sbjct	192	PECWDDVLKSRRIPGHCESLEVACVDNAAGDPPFAEFFFFKCAEHVSGGEKDFAAAPLNLIK	251

Query	231	TNSRNITCITCTDVRSPVLVFCNSRHVICLDCFHLYCVTRLNDRQFVHDPQLGYSLPCV	290
		N +N+ C+ CTDV VLVF C S+HV C+DCF YC +RL +RQF+ P GY+LPC	
Sbjct	252	NNIKNVPLACTDVSDTVLVFFPCASQHVTCIDCFRHYCRSLRGERQFMHPDFGYTLPCP	311

Query	291	AGCPNSLIKELHHFRILGEEQYNRYQQYGAEECVLQMGGVLCPRPGCGAGLLPEPDQRKV	350
		AGC +S I+E+HHF++L E+Y+RYQ++ EE VLQ GGVLC+PGCG GLL EPD RKV	
Sbjct	312	AGCEHSFIEEIHFFKLTLTREEYDRYQRFATEEYVLQAGGVLCPPGCGMGLLVEPDCRKV	371

Query	351	TCEGGNGLGCGFAFCRECKEAYHEGECSAVFE-ASGTTTQAYRVDERAAEQARWEAASKE	409
		TC+ GCG+ FCR C + YH GEC AS T + Y VD A +ARW+ AS	
Sbjct	372	TCQN---GCGYVFCRNCLQGYHIGECLEPGTGASATNSCEYTVDPNRAAEARWEASNV	427

Query	410	TIKKTTKPCPCRHVPVEKNGGCMHMKCPQPQCRLEWCWNCGCEWNRVCMGDHWF	463
		TIK +TKPCP+C P E++GGCMHM C + C EWCW C EW R CMG HWF	
Sbjct	428	TIKVSTKPCPKCRTPTERDGGCMHVMCTRACGFEWCWCQTETWTRDCMGAHWF	481

テーブル形式出力 (BLAST Tabular format)

1. query	2. subject (database)	3. %identity	4. align-len	5. mismatch	6. gap_open	7. q_start	8. q_end	9. s_start	10. s_end	11. evalue	12. bit-score
gi 4758884 ref NP_004553.1	O60260	99.785	465	1	0	1	465	1	465	0.0	964
gi 4758884 ref NP_004553.1	Q9JK66	84.946	465	70	0	1	465	1	465	0.0	830
gi 4758884 ref NP_004553.1	Q9WVS6	83.441	465	76	1	1	465	1	464	0.0	812
gi 4758884 ref NP_004553.1	Q7KTX7	42.194	474	241	8	1	463	30	481	7.05e-128	382
gi 4758884 ref NP_004553.1	E0VIU9	38.445	463	249	9	3	463	32	460	2.53e-113	344
gi 4758884 ref NP_004553.1	Q6T486	27.823	248	139	10	219	453	117	337	5.20e-14	77.8
gi 4758884 ref NP_004553.1	Q84RR0	29.004	231	125	10	231	453	130	329	9.49e-14	77.0
gi 4758884 ref NP_004553.1	Q6PFJ9	26.500	200	119	7	257	453	175	349	9.81e-12	70.5
gi 4758884 ref NP_004553.1	Q9Z1K5	26.500	200	119	7	257	453	203	377	1.01e-11	70.5
gi 4758884 ref NP_004553.1	A2VEA3	26.500	200	119	7	257	453	203	377	1.01e-11	70.5
gi 4758884 ref NP_004553.1	Q9Y4X5	26.500	200	119	7	257	453	205	379	1.07e-11	70.5
gi 4758884 ref NP_004553.1	Q94981	26.538	260	158	11	209	464	105	335	1.34e-11	70.1
gi 4758884 ref NP_004553.1	Q6NW85	26.500	200	119	7	257	453	181	355	1.35e-11	70.1
gi 4758884 ref NP_004553.1	B1H1E4	26.500	200	119	7	257	453	177	351	1.75e-11	69.7
gi 4758884 ref NP_004553.1	Q32NS4	26.066	211	126	8	257	464	177	360	4.50e-11	68.6
gi 4758884 ref NP_004553.1	Q8L829	27.826	230	127	10	232	453	126	324	1.33e-10	67.0
gi 4758884 ref NP_004553.1	Q9UBS8	28.319	226	129	8	256	464	239	448	3.33e-10	65.5
gi 4758884 ref NP_004553.1	Q9Z1K6	23.137	255	157	9	203	453	110	329	4.63e-10	65.1
gi 4758884 ref NP_004553.1	P0C8K8	28.017	232	129	11	230	453	125	326	7.87e-10	64.7
gi 4758884 ref NP_004553.1	Q95376	24.016	254	156	9	203	453	111	330	7.95e-10	64.7
gi 4758884 ref NP_004553.1	Q9SKC3	26.786	224	124	10	235	449	125	317	1.90e-09	63.5
gi 4758884 ref NP_004553.1	Q9JI90	26.009	223	136	7	261	464	237	449	6.37e-09	61.6
gi 4758884 ref NP_004553.1	F1RB95	26.744	172	104	5	306	462	266	430	8.54e-09	61.2
gi 4758884 ref NP_004553.1	Q05120	33.333	72	48	0	1	72	1	72	1.27e-08	55.5

この出力形式は柔軟に変更できる。

コマンドライン BLAST

データベースを自前で用意する

```
makeblastdb -in 配列ファイル -dbtype {prot|nucl} -out データベース名
```

- 配列ファイル human.fa にインデックスをつけ、humanというデータベースを作る。
% makeblastdb -in human.fa -dbtype prot -parse_seqids -out human
- 作成したデータベースに対して、mouse0.fa をクエリとして検索を実行する。E-valueの閾値を0.001とし(-evalue)、クエリ当たり上位10個までのヒットを出力(-max_target_seqs)。
% blastp -query mouse0.fa -db human -num_threads 4
-outfmt 6 -evalue 0.001 -max_target_seqs 10 > mouse0_human.out